ROGER WONG SIE YANG

+65 8713 4620 | rogerwong20011021@gmail.com | Singapore | LinkedIn | Portfolio

SUMMARY

Full Stack Software Engineer with 2+ years of production application experience across fintech, enterprise systems, and cybersecurity software. Building end-to-end AI-assisted compliance workflows incorporating LLM integration, agent orchestration, and full-stack application development. Experienced with Java Spring Boot, Python, TypeScript, Next.js, Angular, RESTful APIs, SQL databases, AWS, and CI/CD. Bachelor of Computer Science (Artificial Intelligence), CGPA 3.92; seeking to contribute to scalable enterprise AI applications and operational automation.

EDUCATION

Universiti Malaya **2020 - 2024**
- Bachelor of Computer Science (Major in Artificial Intelligence) (CGPA: 3.92)

Kolej Matrikulasi Labuan **2018 - 2020**
- Matriculation Programme (CGPA: 4.00)

Sekolah Menengah Kebangsaan Sacred Heart **2013 - 2018**
- Malaysian Certificate of Education (SPM: 5A+, 3A, 2A-)

WORK EXPERIENCE

Athena Dynamics Pte Ltd | Singapore *Subsidiary of BH Global (SGX: BQN)* **Oct 2026 - Present**

Software Engineer
- Building a full-stack AI-assisted maritime compliance application, connecting user-facing review workflows, backend services, and LLM capabilities into an end-to-end enterprise solution.
- Developing agent-oriented workflows and LLM orchestration harnesses for document analysis, contextual question answering, and structured compliance review.
- Designing reusable AI integration components for prompts, tool interactions, and review workflows, with attention to traceability, secure data handling, and maintainability.
- Developing full-stack application services using TypeScript/Next.js, REST APIs, and PostgreSQL to support structured compliance data, review records, and AI-assisted workflows.
- Supporting AWS-hosted application architecture and deployment for integrated web, database, and LLM-enabled services, with attention to security, scalability, and maintainability.

iFAST Global Hub AI | Kuala Lumpur *iFAST Corporation (SGX: AIY)* **Apr 2024 - Sep 2026**

IT Specialist (Full Stack Software Engineer)
- Spearheaded enhancement of fintech mobile modules using Flutter across the FSMOne investment platform (HK/MY/SG) and a UK SME digital banking platform, improving transaction latency and optimizing trading, payment, and portfolio management workflows in production environments.
- Contributed to full-stack development of a UK digital banking platform for SME and NBFI clients, implementing secure authentication and payment workflows that improved onboarding turnaround time.
- Implemented secure domestic transfer and FX payment services through RESTful APIs and microservices architecture, increasing successful transaction completion rates by 22%.
- Redesigned eKYC and compliance back-office workflows to resolve manual verification bottlenecks, substantially improving operational efficiency and strengthening fraud detection accuracy.
- Engineered backend services using Java Spring Boot with PostgreSQL and MySQL to stabilize settlement processing, decreasing system downtime incidents and improving overall platform reliability.
- Implemented automated CI pipelines using GitLab CI/CD, with Argo CD used to manage and monitor production deployments, improving release management efficiency and deployment visibility.
- Developed Angular-based analytics dashboards to centralize operational monitoring, enabling real-time reporting visibility and improving internal decision-making efficiency by 30%.

Seamless X Sdn Bhd | Kuala Lumpur **Jul 2023 - Oct 2023**

Full Stack Software Engineer (Part Time)
- Architected an administrative web dashboard using .NET MVC and MySQL to consolidate service operations, enhancing data accuracy and improving workflow efficiency by 30%.
- Optimized backend queries and validated RESTful endpoints using Postman to resolve recurring system inconsistencies, reducing reported technical issues and enhancing system stability.
- Collaborated with stakeholders to gather functional requirements and deliver system enhancements under Agile methodology, accelerating feature rollout timelines and improving cross-functional coordination.
- Implemented role-based access control and authentication mechanisms to strengthen system security, improving data protection standards and ensuring controlled access across multiple internal service modules.

Fusionex Sdn Bhd | Petaling Jaya **Jul 2022 - Dec 2022**

Software Engineer Intern

- Built a cross-platform Flutter e-commerce application using BLoC architecture to address user experience gaps, achieving over 10,000 active users and maintaining a 4.5-star rating.
- Developed a warehouse management system using Vue.js and .NET Core with optimized MongoDB indexing, strengthening inventory tracking accuracy for over 2,000 stock items.
- Implemented RESTful APIs and integrated frontend components to automate inventory synchronization processes, reducing manual tracking errors and improving system reliability before production rollout.

PROJECTS

MEMOSA | Universiti Malaya **Mar 2024**

- Performed cloud migration from AWS Singapore to AWS Malaysia region to comply with data security and residency requirements.
- Designed and deployed a production-grade AWS infrastructure for Memosa using VPC, ECS (Fargate), API Gateway, Network Load Balancer, and Aurora MySQL, enabling scalable and secure microservices architecture.

RESPIRE | Universiti Malaya **Jan 2024**

- Developed a cross-platform air quality forecasting application using Flutter, Golang, Firebase, and MongoDB, integrating real-time environmental models to visualize PM2.5, NO2, and O3 data for Klang users.

Advance Care Planning ACP | Universiti Malaya **Apr 2023**

- Built a mobile healthcare application using Flutter and Next.js that enabled users to create structured medical directives through interactive modules and secure cloud data storage.

VOLUNTEER & LEADERSHIP

Director of Programming League National | Universiti Malaya **Jan 2022 - Jun 2022**

- Led a national-level programming competition involving over 300 participants across Malaysia, managed a 70-member organizing team, coordinated weekly progress meetings, and ensured successful end-to-end event execution.

SKILLS & LANGUAGES

Technical Skills

- Programming Languages: Java, Python, JavaScript, TypeScript, Dart
- Mobile Development: Flutter, Swift, Kotlin, Android Studio, Xcode
- Frontend Technologies: Angular, React, Vue.js, Next.js, HTML, CSS, jQuery
- Backend Technologies: Java Spring Boot, .NET Core, Golang, RESTful APIs
- Database Management: MySQL, PostgreSQL, MongoDB, Firebase Firestore
- AI & Application Concepts: LLM Integration, Agentic Workflows, Prompt Orchestration, AI-Assisted Document Review
- DevOps and Tools: AWS, Git, GitLab CI/CD, Argo CD, Firebase App Distribution, TestFlight, Postman
- Architecture and Concepts: Microservices, MVC, BLoC Architecture, Agile Methodology

Soft Skills

- Problem Solving, Analytical Thinking, Team Leadership, Communication, Critical Thinking, Time Management, Adaptability, Collaboration

Languages

- English (Fluent), Mandarin (Fluent), Malay (Basic)

REFERENCES

- Available upon request